

WIRELESS NETWORKING

DAY – 3 INTRODUCTION TO IP

ADDRESS THEORY

TASK – 1 IPV4 and IPV6.

Feature	IPv4	IPv6
Address Length & Space	Uses a 32-bit address space, allowing for approximately 4.3 billion unique IP addresses.	Uses a 128-bit address space, allowing for approximately 3.4×10^{38} (340 undecillion) unique IP addresses.
Address Format	Written in numeric decimal notation, separated by dots (e.g., 192.168.1.1).	Written in hexadecimal notation, separated by colons (e.g., 2001:0db8:85a3:0000:0000:8a2e:0370:7334).
Address Configuration	Supports manual assignment or automatic dynamic assignment via DHCP (Dynamic Host Configuration Protocol).	Supports Stateless Address Autoconfiguration (SLAAC) without a DHCP server, as well as DHCPv6.

Header Size & Complexity	Variable length header size (typically 20 to 60 bytes) with optional fields, requiring more router processing.	Fixed header size (40 bytes) with a simplified structure, enabling faster and more efficient packet routing.
Packet Fragmentation	Performed by both sending hosts and intermediate routers along the path.	Performed exclusively by the sending host (end-to-end); routers do not fragment IPv6 packets.
Security (IPsec)	IPsec support is optional and typically relies on third-party security implementations.	IPsec (Internet Protocol Security) support is built into the protocol specifications.

Task – 2 Assign static IPv4 address.

- Here is a valid static IPv4 network configuration for your 4 devices on the **192.168.10.0/24** network.
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- **Network Parameters**
- **Network Address:** 192.168.10.0/24
- **Subnet Mask:** 255.255.255.0
- **Default Gateway (Router IP):** 192.168.10.1
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- **Device Configurations**
- **Laptop**
- **IP Address:** 192.168.10.10

- **Subnet Mask:** 255.255.255.0
- **Default Gateway:** 192.168.10.1
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- **Phone**
- **IP Address:** 192.168.10.11
- **Subnet Mask:** 255.255.255.0
- **Default Gateway:** 192.168.10.1
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- **Smart TV**
- **IP Address:** 192.168.10.12
- **Subnet Mask:** 255.255.255.0
- **Default Gateway:** 192.168.10.1
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- **Printer**
- **IP Address:** 192.168.10.13
- **Subnet Mask:** 255.255.255.0
- **Default Gateway:** 192.168.10.1
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Task – 3 Calculating the valid subnets.

- 1. Subnet & Host Calculations
- Network Class: Class C (Default mask: /24 or 255.255.255.0)
- Subnet Mask: /27 = 255.255.255.224
- In binary: 11111111.11111111.11111111.11100000
- Subnet Bits Borrowed (\$s\$): $27 - 24 = 3 \text{ bits}$
- Host Bits Remaining (\$h\$): $32 - 27 = 5 \text{ bits}$
- *Calculations:*
- Number of Subnets: $2^s = 2^3 = \mathbf{8 \text{ valid subnets}}$
- Total IPs per Subnet: $2^h = 2^5 = 32$
- Usable Hosts per Subnet: $2^h - 2 = 32 - 2 = \mathbf{30 \text{ usable hosts}}$
- Block Size (Increment): $256 - 224 = \mathbf{32}$
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- 2. First Three Subnet Ranges
- *Subnet 1*
- Network Address: 192.168.1.0

- Usable Host IP Range: 192.168.1.1 to 192.168.1.30
- Broadcast Address: 192.168.1.31
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- *Subnet 2*
- Network Address: 192.168.1.32
- Usable Host IP Range: 192.168.1.33 to 192.168.1.62
- Broadcast Address: 192.168.1.63
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- *Subnet 3*
- Network Address: 192.168.1.64
- Usable Host IP Range: 192.168.1.65 to 192.168.1.94
- Broadcast Address: 192.168.1.95

Task –4 Allocating the network address, broadcast address.

Property	Value
IP Address	10.0.5.17
Subnet Mask	255.255.255.0 (/24)
Network Address	10.0.5.0
Broadcast Address	10.0.5.255
Valid Host Range	10.0.5.1 – 10.0.5.254
Total Usable Hosts	254

Task – 5 Using the subnet tools calculator.

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IPv4 Subnet Calculator

Result



IP Address:	10.0.0.0
Network Address:	10.0.0.0
Usable Host IP Range:	10.0.0.1 - 10.0.0.2
Broadcast Address:	10.0.0.3
Total Number of Hosts:	4
Number of Usable Hosts:	2
Subnet Mask:	255.255.255.252
Wildcard Mask:	0.0.0.3
Binary Subnet Mask:	11111111.11111111.11111111.11111100
IP Class:	C